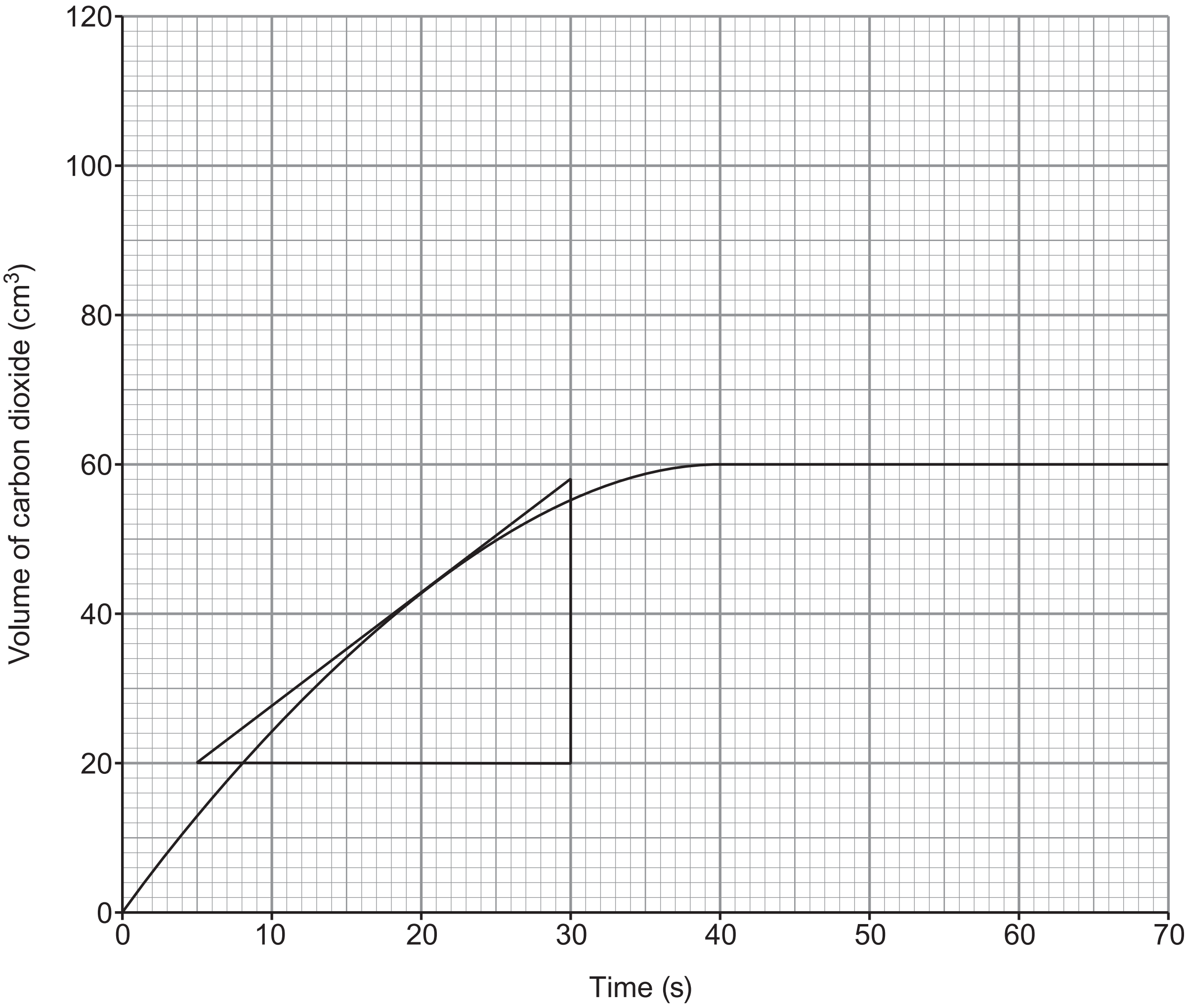
11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol / dm^3 hydrochloric acid.



(a) Use the tangent drawn to calculate the rate of the reaction at 20 s. [2]

Rate = ... cm^3 / s

Examiner only

(b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]

(ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

....

....

....

....

....

....

6

END OF PAPER